

Proposal: Solar + Battery Microgrid for Meadow Hall Schools

Prepared for: Meadow Hall Schools

Objective: Cut electricity spend, improve reliability, and stabilize power quality by powering the campus through a solar + battery inverter-backed microgrid.

1) Executive Summary (The Offer)

We propose a **~250 kWp Solar PV + 300 kWh Battery + Site Inverter System** designed to:

- Supply a large portion of daytime energy directly from solar
- Use batteries to smooth solar variability (clouds/irradiance dips) and support short-duration autonomy
- Feed the site via an inverter-controlled bus for better power quality and controllability

Financial headline (at ~~£~~300/kWh):

- Estimated annual savings: ~~~£94m~~ **~£113m net/year** (range depends on solar yield and uptime)
- Simple payback: **~2.4–2.9 years** (most-likely **~2.6 years**)

2) System Size (Based on Meadow Hall Peak Demand)

From historical data, Meadow Hall's measured peak demand reaches approximately **443 kW**.

Proposed system (base package):

- **Solar PV:** ~250 kWp ($\approx$ 455 panels)
- **Battery:** 300 kWh (LFP)
- **Inverter system (site PCS/inverter):** budgeted at ~~£~~90m (final kW/kVA rating to be confirmed during site survey and motor-start assessment)
- **Cables/installation:** ~~£~~20m

Note: The PV and battery reduce imported energy; they do not “create” energy beyond what solar provides. Battery mainly improves reliability and enables solar utilization during variability.

3) Energy & Savings (₦300/kWh)

3.1 Expected PV production (Lagos yield)

Assuming ~**250 kWp** installed and typical Lagos yield of **3.8–4.6 kWh/kWp/day**:

- Annual PV energy: ~**347,000 – 420,000 kWh/year**

3.2 Savings

At **₦300/kWh** avoided cost:

- Gross savings: **₦104m – ₦126m / year**
- Net savings (≈10% allowance for losses, soiling, downtime): **₦94m – ₦113m / year**

4) ROI / Payback (Using Your CAPEX)

4.1 Investment (CAPEX)

Item	Quantity / Basis	Amount (₦)
Inverters	Lump sum	90,000,000
Panels	455 × ₦150,000	68,250,000
Batteries	300 kWh	90,000,000
Cables/Installation	Lump sum	20,000,000
Total CAPEX		268,250,000

4.2 Payback

- Payback = **₦268.25m / annual net savings**

- Estimated payback: **~2.4 – 2.9 years** (most-likely **~2.6 years**)

5) Battery “1-Hour Autonomy” (Clarified)

The **300 kWh** battery is excellent for **solar degradation ride-through** and short-duration backup.

- At **full peak load (443 kW)**, 300 kWh provides **~0.7 hours theoretical** (less after usable DoD and losses).
- If Meadow Hall requires **a full 1 hour at the 443 kW peak**, we recommend upgrading to **~800 kWh** battery capacity.

Recommendation: Confirm the required autonomy definition:

- **Ride-through during solar dips** (most common) → **300 kWh is typically sufficient**
- **Full-site 1-hour autonomy at peak** → **upgrade battery capacity**

6) What Meadow Hall Gets (Value Beyond Savings)

- **Lower electricity spend** with predictable solar generation
- **Improved uptime** during grid instability and solar variability
- **Cleaner, quieter operations** (reduced generator runtime)
- **Better power quality** (stable voltage/frequency under inverter control)
- **Remote monitoring** for performance assurance and rapid fault response

7) Implementation Plan (High Level)

1. **Site survey & data validation** (loads, motor starts, distribution voltage, protection)
2. **Engineering design** (single line diagram, protection coordination, inverter sizing)
3. **Procurement** (modules, inverters, batteries, switchgear)
4. **Installation & commissioning** (testing, handover, operator training)

8) Key Assumptions (For This Financial Model)

- Solar yield basis: **3.8–4.6 kWh/kWp/day**
- Energy value: **₺300/kWh**
- Net savings include **~10%** allowance for typical real-world losses
- Final inverter rating and electrical scope depend on site survey (motors, voltage level, transformer capacity, protection upgrades)

9) Next Steps

1. Confirm panel wattage (e.g., **550 W** assumed for the 455-panel count)
2. Confirm autonomy requirement (ride-through vs full 1-hour at 443 kW peak)
3. Schedule site survey and finalize engineering + fixed-price scope

We're ready to proceed with a site survey and detailed design to lock final performance and guarantees.